



Figure 1: The test performance on the Concrete and Housing datasets for the IDent method and the PIDent method when the number of training examples for partial-label regression increases.

slightly better performance than the IDent method, which implies that the progressive identification strategy (i.e., soft weights) could be more promising than the direct identification strategy (i.e., hard pseudo-labeling). We may expect more significant improvements of the PIDent method over the IDent method with a better designed weighting function. 5) The AVGL methods are generally better than the AVGV methods, which implies that training with average loss is generally better than average value. Besides, MAE and Huber obviously outperform MSE, which shows that the robustness of MAE and Huber still holds in the PLR task.

Performance of Increasing PLR Training Data. As we showed in Theorem 2 and Theorem 4, the models learned by our proposed IDent method and the PIDent method could converge to the optimal model learned from fully labeled data when the number of PLR training examples approaches to infinity. Therefore, the performance of the two methods is expected to be improved if more PLR training examples are provided. To empirically validate such a theoretical finding, we further conduct experiments on the Concrete and Housing datasets by changing the fraction of PLR training examples (100% means that we use all the PLR training examples in the training set). The experimental performance of the IDent method and the PIDent method when we increase the PLR training examples is provided in Figure 1. As shown in Figure 1, the test loss of the two methods generally decreases when more PLR training examples are used for model training. This observation is clearly in accordance with our theoretical analyses in Theorem 2 and Theorem 4, because the learned model would be closer to the optimal model as more PLR training examples are provided.

Conclusion

In this paper, we investigated a novel weakly supervised learning setting called partial-label regression, which is a variant of partial-label learning focusing on the regression task. To solve this problem, we first proposed a simple baseline method that takes the average loss incurred by candidate labels as the predictive loss. To overcome the drawback of this baseline method, we proposed an identification method that takes the least loss incurred by candidate labels as the predictive loss. We further propose a progressive identification method to differentiate candidate labels using

progressively updated weights. We proved the model consistency of the latter two methods, which indicates that learned model can converge to the optimal model learned with fully labeled data. Finally, we conducted extensive experiments to demonstrate the effectiveness of our proposed methods. We expect that our first study with simple yet theoretically grounded methods for partial-label regression could inspire more research works on this new task.

We only used a uniform distribution to generate real-valued candidate label sets due to the page limit in the experiments. It would be interesting to further empirically investigate the performance of our proposed methods with other types of generation distributions. Besides, since our proposed methods can be compatible with any models, optimizers, and loss functions, they would be suited for dealing with large-scale regression datasets. Therefore, another promising direction is to apply our proposed methods to solve large-scale problems encountered in real-world application domains such as computer vision (?) and natural language processing (?). In addition, the performance of the PIDent method heavily relies on the designed weighting function. Hence how to properly design an effective weighting function for the PIDent method would be an interesting direction to further improve this method.

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